Experimental Investigation and Thermodynamic Optimization of the Y-Cu Binary System

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The Y-Cu binary system was experimentally investigated by DTA and XRD. The DTA results provide evidence for the possible existence of a high-temperature phase transformation in the YCu_2 compound. The XRD results confirm a range of solubility for the YCu_6 phase. A thermodynamic description for all the phases was obtained by taking into account all experimental data in the literature. A new phase diagram is proposed.

Introduction

The Y-Cu, Ba-Cu, and Ba-Y binary systems form the ternary metallic subsystem of the quaternary Y-Ba-Cu-O system, for which a thermodynamic description is searched. This quaternary system contains the superconducting phase $YBa_2Cu_3O_{7-x}$. For a thermodynamic description of the quaternary system, reliable experimental and thermodynamic descriptions of the binary subsystems must be available.

The Ba-Cu binary system was recently experimentally reinvestigated, and the equilibrium temperature of the peritectic formation of the $BaCu_{13}$ compound was found to be much higher than the temperatures previously reported in the literature. This experimental result was taken into account in the new thermodynamic description for this binary system [93Kon].

The only information available on the Ba-Y system is found in a theoretical survey of Y-alkali metal systems by [86Dju]. No compounds were reported, and the liquid phase has a miscibility gap with a high unknown critical temperature. This temperature can be estimated through experimental investigations on the liquid of the Ba-Cu-Y ternary system [94Kon].

The Y-Cu binary system was previously assessed by [81Cha], and more recently the phase diagram was calculated from an optimized thermodynamic description by [90Ita] using the experimental data obtained by [89Qi]. According to [92Oka], who accepted the Itagaki et al. [90Ita] calculated diagram, the system presents five intermetallic compounds: YCu, YCu_2, Y_2Cu_7, YCu_4, and YCu_6 (denoted YCu_7 by [71Bus]). A range of solubility is attributed to YCu_6. Apart from YCu_6 and Y_2Cu_7, all phases melt congruently. There is still some controversy in the literature with regard to the temperatures of the binary invariant reactions. In addition, the ranges of solubility reported in the literature for the YCu_4 and YCu_6 phases [81Cha], and calculated by [90Ita] for YCu_6, have not yet been determined quantitatively.

These facts motivated the present experimental investigation and the subsequent thermodynamic optimization of the Y-Cu system.

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Experimental

Experimental Procedure

The DTA results in this study were obtained using pyrolytic boron nitride (PBN) crucibles, which had 6 mm length, 4 mm inner diameter, ½ mm thick wall, and ½ mm thick flat lid (Ringsdorff GmbH). The sample mass was ~180 mg, which was optimal for our study. The starting materials were Cu chips or rod (99.999 wt.% pure) and Y chips (99.9 wt.% pure). Yttrium powder (99.9 wt.% pure) was used in the beginning of the study, but this resulted in excessive Y loss.

The small Cu pieces were freshly cleaned in a 10% HCl solution to remove the surface oxide, washed in methanol, rinsed in acetone, and dried. The Cu and Y were weighed in air (±0.1 mg) and for melting placed into a PBN crucible twice as long as a DTA crucible (the starting materials did not fit into the DTA crucible). The melting crucible was placed into a silica tube, a PBN lid and silica spacer were placed on top, and a small amount of Ti-getter was added. After repeatedly evacuating the sample to $<10^{-4}$ bar and flushing with argon, the silica

tube was sealed and the Ti-getter was heated with the torch for several minutes to remove all the remaining oxygen from the gaseous phase. The sample was then isolated from the Ti-getter by sealing the silica tube. The samples were premelted at 1393 K for ~½ h, water quenched, and annealed for approximately 12 h at 1123 K. The samples could be easily removed from the melting crucibles because they formed small spheres with diameters less than the crucible. The sample normally had a small amount of surface discoloration, which could be easily removed with fine SiC paper.

The samples for DTA analysis were placed into the smaller PBN crucible, which was held in an alumina support crucible with a hole in the bottom to allow direct contact between the thermocouple and the PBN crucible. A small Pt sphere was used as the DTA reference material and was placed in similar PBN and alumina crucibles. Then the DTA furnace was repeatedly evacuated using a mechanical pump and refilled with argon (99.998 wt.% pure). In order to reduce the amount of oxygen contamination, the argon was first passed through an oxygen titration unit. This unit consisted of a solid-state electrolyte tube (calcium-stabilized zirconia), which had Pt electrodes painted both inside and outside of the active zone of the tube. The active zone was held in a small furnace at 1073 K, and oxygen could be titrated out of the argon by placing a voltage across the electrodes (0.8 to 1 V).

In order to calibrate the DTA system, pure Sb, Ag, Cu, and Au standards were analyzed using the same Pt reference, PBN, and alumina crucibles. The calibration offset, with respect to the tabulated standard emf of Pt/Pt-Rh thermocouples, varied from 7 to 0 K among the different DTA sample holders used in the course of this study.

The DTA results were obtained using heating and cooling rates of 0.067 K/s and an argon flow rate of 0.83 mL/s. The reaction onset temperature was chosen as the intersection of the extrapolated baseline with the tangent drawn to the greatest slope of the leading edge of the DTA peak. Since the same analysis was used for the calibration standards, most side effects should be compensated.

In addition to the DTA analysis, powder XRD work was performed on several samples in the composition range 80 to 90 at.% Cu. The XRD samples were prepared in the same way as the DTA samples, except that they were annealed either at 1123 K for 7 days or at 973 K for 18 days (sample N21, which was made subsequent to the other samples, was for only 6 days). After annealing, the samples were milled to powder in a zirconia ball mill and analyzed by a diffractometer using Cu-K_α radiation and a 2Θ step size of 0.04°.

Experimental Results

Numerous difficulties were encountered during the XRD and metallographic investigations. Initial work using bulk alloys (0.5 to 1 g) was often confusing and ambiguous. Satisfactory results were obtained using only very small spheres (180 mg), which were rapidly quenched after melting and annealed for long times at high temperatures.

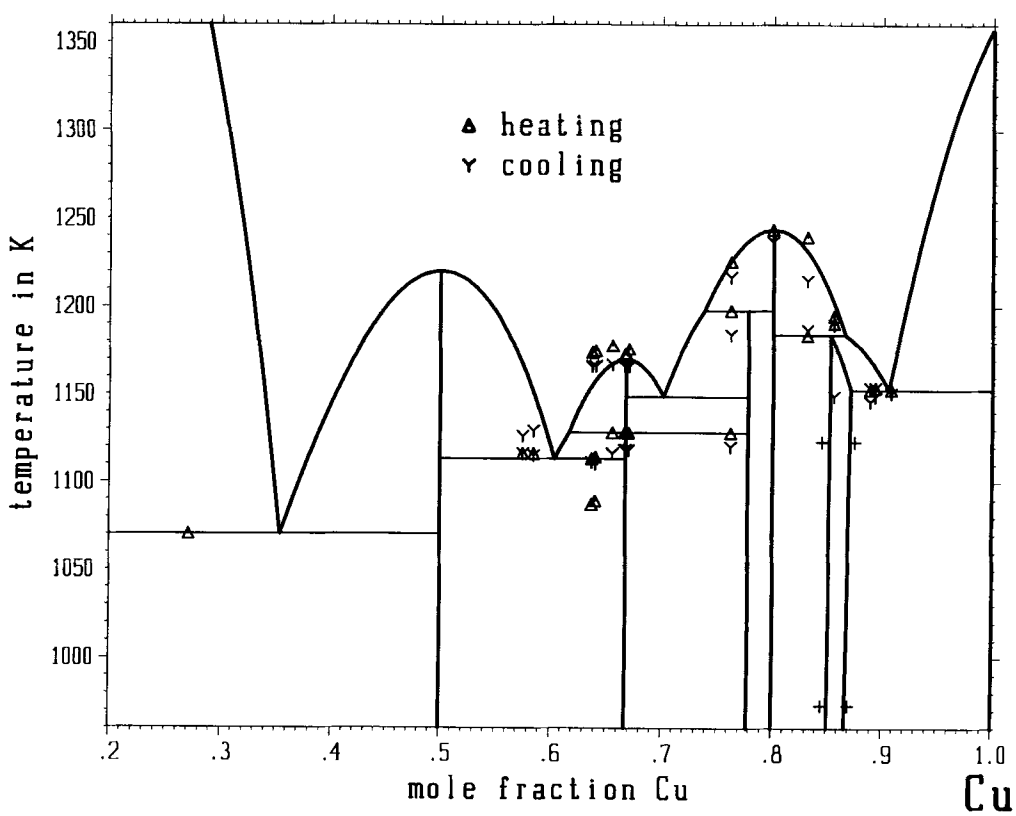

Fig.1 DTA experimental results on heating and cooling, plotted in the Y-Cu phase diagram.

Differential Thermal Analysis

The DTA results obtained on heating and cooling are plotted in the calculated Y-Cu phase diagram in Fig. 1, where each data point is the average of several heating and cooling cycles. The invariant equilibrium temperatures obtained agree well with most of the experimental results from [89Qi]. However, since the mass of the samples used in this study was small (180 mg), a small loss of Y will have a significant impact on the sample composition. Therefore in the subsequent optimization, only the invariant equilibrium temperatures, but not the liquidus data, were used.

Examples of DTA heating and cooling curves obtained near the YCu_2 composition are shown in Fig. 2. The real compositions may be slightly shifted to the more Cu-rich side from the nominal ones given in Fig. 2, but no corresponding quantitative data can be given.

X-Ray Diffraction

In Fig. 3, XRD spectra for samples annealed at 1123 K containing 80.04 at.% Cu (YCu_4) and 86.19 at.% Cu (sample N15) are shown. Many attempts were made to obtain a clearer XRD spectrum for YCu_4, but these always resulted in a similar diffuse pattern. It should be noted that the YCu_4 spectrum had a step time (measurement time at each angle) five times longer than the YCu_6 spectrum shown in Fig. 3. Thus the background and noise levels are much higher because both spectra are raw data without background removal, smoothing, or K_{α_2} subtraction. The small broad peak near 20° 2Θ in both spectra (much smaller in spectrum *b*) comes from the adhesive film used to attach the powder to the sample holder.

A summary of the XRD phase analysis and lattice parameter determination is given in Table 1. For the two-phase YCu_6+YCu_4 samples, a distinct YCu_4 pattern cannot be observed for compositions of >82 at.% Cu. Primary evidence for the presence of the YCu_4 phase is the increase of the background level and of the peak widths between 40 and 45° 2Θ. In sample N17 at 84.27 at.% Cu, evidence of YCu_4 could not be detected; thus an additional sample, N21, was prepared with a lower Cu-content of 83.22 at.% Cu and a shorter annealing time of 6 days.

Discussion

The results of the DTA study are consistent with the experimental work of [89Qi]. The DTA curves obtained near the YCu_2 compound (see Fig. 2) clearly show two effects, the intensity ratio of which is virtually independent of composition. The lower effect cannot be explained as being due to the eutectic $L \leftrightarrow YCu_2 + Y_2Cu_7$ because in this case its intensity should be proportional to the distance from the compound composition disappearing totally at the YCu_2 composition. The only possible explanation of this DTA effect is a solid state transformation $YCu_2(h) \leftrightarrow YCu_2(r)$ (hexagonal to rhombohedral). Some evidence of this transformation is reported by [63Kej], who always observed twinning in the YCu_2 phase. The $RECu_2$ compounds are isostructural with $CeCu_2$; only $LaCu_2$ has the hexagonal AlB_2 structure. These structures are closely related. Attempts to find a transformation of $LaCu_2$ to a low-temperature $CeCu_2$-type phase or of YCu_2 to a high-temperature, hexagonal AlB_2-type phase were reported by [63Sto]. They did not find a transformation, but left open the question as to whether by selecting other annealing temperatures a transformation would be found. Compared with the effect shown in Fig. 2, the temperature chosen by [63Kej] was too low to enable the $YCu_2(h)$ formation. Note that the "liquidus" temperatures (second peak on heating) for all three samples are almost the same, 1166 K upon cooling, indicating a very flat liquidus at YCu_2. This liquidus temperature remains constant for each

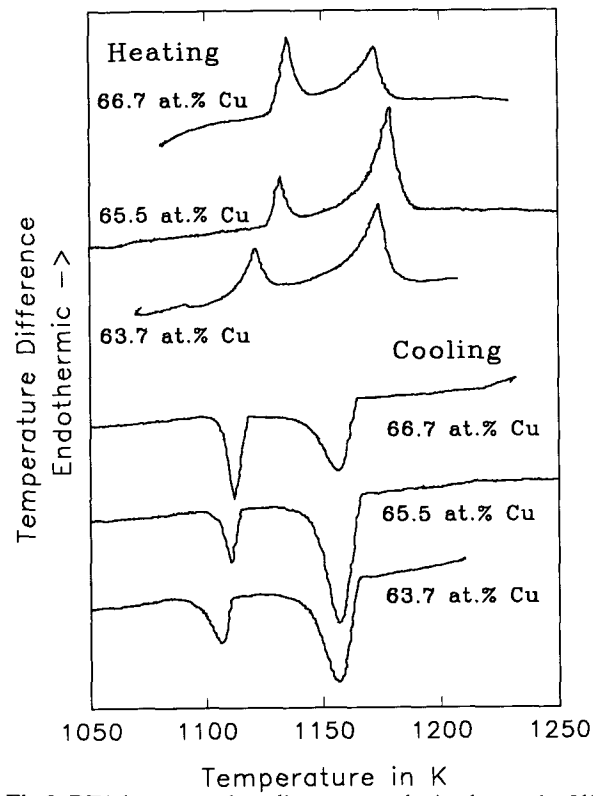

Fig.2 DTA heating and cooling curves obtained near the YCu_2 composition.

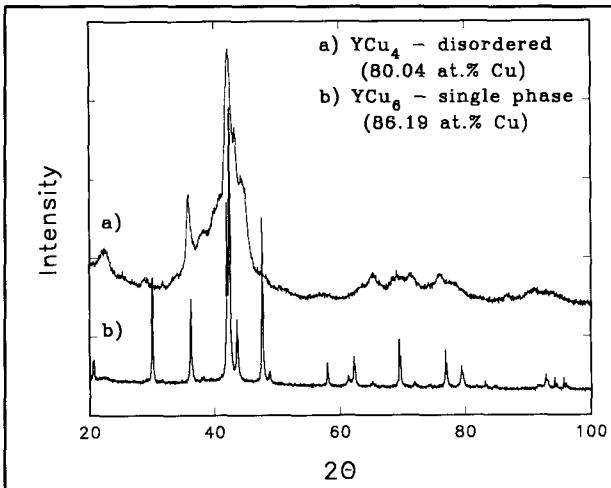

Fig.3 XRD spectra obtained from YCu_4 and YCu_6 at 1123 K. The spectra are raw data, with spectrum *a* having a step time 5 times longer than *b*.

Table 1 Summary of Conclusive XRD Data for Y-Cu Samples near the "YCu_6" Composition

Sample	Composition, at.% Cu	Annealing temperature °C	Observed phases	Hexagonal YCu_6 lattice parameters, nm		
				a	*c*	*c/a*
N12	83.41	850	$YCu_6 + YCu_4$	0.4985(3)	0.4122(4)	0.827
N15	86.19	850	YCu_6	0.4960(1)	0.4142(1)	0.835
N10	89.65	850	$YCu_6 + Cu$	0.4940(2)	0.4152(2)	0.840
N21(a)	83.22	700	$YCu_6 + YCu_4$	0.4982(14)	0.4122(21)	0.827
N17	84.27	700	YCu_6	0.4985(3)	0.4124(4)	0.827
N16	88.10	700	$YCu_6 + Cu$	0.4949(4)	0.4150(5)	0.839

(a) Sample N21 was annealed only 6 days.

sample, even after several heating and cooling cycles. Similarly, [89Qi] obtained a liquidus temperature of 1165±5 K for a Y-Cu sample containing 65.9 at.% Cu and a melting temperature of 1170±5 K for YCu_2.

In previous XRD studies of the Cu-rich region, only the phases denoted YCu_5 [59Wer, 61Dwi, 71Bus, 82Chu] and YCu_7 [71Bus] were reported. However, considering the difficulties experienced in this study with YCu_6+YCu_4 two-phase samples, the existence of the YCu_4 was probably overlooked in the previous studies. The *c/a* ratio obtained in this study for the Y-rich YCu_6 phase is quite similar to that obtained for the YCu_5 phase [59Wer, 61Dwi, 82Chu] with the exception of the value from [71Bus], who obtained $c/a = 0.812$ for a splat-cooled sample. It is likely that [71Bus] was able to produce a metastable YCu_5 phase under such rapid cooling conditions. The results from this study for the Cu-rich YCu_6 phase are similar to those obtained by [71Bus] for a phase called YCu_7, but stated to exist at slightly lower Cu concentration.

The homogeneity range of YCu_6 can be deduced from the lattice parameters. Plotting the homogeneous samples N15 and N17 together with the splat-cooled metastable sample YCu_5 of [71Bus] in a lattice parameter-composition plot results in a straight line for *a* and a less perfect straight line for *c*. Comparing the lattice parameters of the two-phase $YCu_4 + YCu_6$ samples, N12 and N21, gives the limit at the Y-rich side of 84.5 ± 0.5 at.% Cu. The lattice parameter of the N10 and N16 samples yield for the Cu side boundary 87.0 ± 0.5 at.% Cu. [71Bus] reported the existence of a $CaCu_5$-type structure at a higher Cu content than YCu_5. The formula YCu_7 given by him does not represent a stoichiometric composition, but the maximum Cu content. The excess Cu content was shown to be due to a random substitution of Y atoms by Cu pairs. Such pairs exist in an ordered arrangement in the *hP*38,Th_2Ni_{17}-type structure, which is related to the *hP*6,$CaCu_5$-type structure replacing 2 of 6 Th atoms by Ni pairs [64Sch] in a unit cell composed of 6 $CaCu_5$-type cells ($a = \sqrt{3}a'$, $c = 2c'$). From this point of view, the YCu_6 phase is clearly the one with the $CaCu_5$-type structure. [61Dom] reported *d* values for the YCu_4 phase; those values agree well with the values reported by other authors and found here for the Y-rich YCu_6 phase.

The fact that no X-ray pattern with sharp lines can be obtained for YCu_4 suggests that the crystal structure of this phase has some defects destroying the periodicity, which occurs, for example, for the Guinier-Preston zones. The X-ray spectrum of YCu_4 has some similarities with that of YCu_6, indicating the possibility that YCu_4 has the $CaCu_5$-type structure with ran-

Table 2 Summary of Phase Diagram Investigations

Invariant equilibrium	Temperature, K	Values in optimization	Reference
$L \leftrightarrow Y + YCu$	1055	No	[61Hae]
	1033 ± 15	No	[61Dom]
	1070 ± 3	Yes	[89Qi]
	1070	...	This work (calc.)
$L \leftrightarrow YCu$	1225	No	[61Hae]
	1208 ± 15	No	[61Dom]
	1220 ± 5	Yes	[89Qi]
	1220	...	This work (calc.)
$L \leftrightarrow YCu_2(r) + YCu$	1109	No	[61Hae]
	1103 ± 15	No	[61Don]
	1110 ± 7	Yes	[89Qi]
	1116 ± 5	Yes	This work (exp.)
	1113	...	This work (calc.)
$YCu_2(r) \leftrightarrow YCu_2(h)$	1128 ± 5	Yes	This work (exp.)
	1128	...	This work (calc.)
$L \leftrightarrow YCu_2(h)$	1168	No	[61Hae]
	1208 ± 15	No	[61Dom]
	1170 ± 5	No	[89Qi]
	1170 ± 5	Yes	This work (exp.)
	1170	...	This work (calc.)
$L \leftrightarrow Y_2Cu_7 + YCu_2$	1135(a)	No	[61Hae]
	1113(a)	No	[61Dom]
	1127 ± 5	No	[89Qi]
	1149	...	This work (calc.)
$L + YCu_4 \leftrightarrow Y_2Cu_7$	1193	No	[66Bea]
	1198	Yes	This work (exp.)
	1198	...	This work (calc.)
$L \leftrightarrow YCu_4$	1258 ± 15	No	[61Dom]
	1248 ± 5	No	[66Bea]
	1240 ± 5	No	[89Qi]
	1244 ± 5	Yes	This work (exp.)
	1244	...	This work (calc.)
$L + YCu_4 \leftrightarrow YCu_6$	1183 ± 10	No	[66Bea]
	1203 ± 20	No	[61Dom]
	1184 ± 5	Yes	This work (exp.)
	1184	...	This work (calc.)
$L \leftrightarrow (Cu) + YCu_6$	1123	No	[57Daa]
	1139	No	[61Hae]
	1163 ± 10	No	[61Dom]
	1133 ± 2.5	No	[75Fed]
	1155 ± 5	No	[89Qi]
	1153 ± 2	Yes	This work (exp.)
	1153	..	This work (calc.)

Note: The invariant equilibria calculated after the optimization are labeled by (calc.). **(a)** Y_2Cu_7 not found.

Table 3 Summary of the Thermodynamic Experimental Investigations

Reference	Experimental method	Function	Cu mole fraction	Temperature, K	Values in optimization
[84Wat]	Reaction calorimetry	Enthalpy of mixing	0 to 0.80	1373	Yes
	Drop calorimetry	Enthalpy of formation	0.50	1373	Yes
			0.6667	1373	Yes
			0.80	1373	Yes
[89Qi]	Drop calorimetry	Heat content	0.33 to 0.907	850 to 1300	Yes
[89Yan]	Electromotive force	μ_Y	0.6667	973 to 1103	No
			0.7778	973 to 1103	No
			0.85	973 to 1103	No
			100	973 to 1103	No

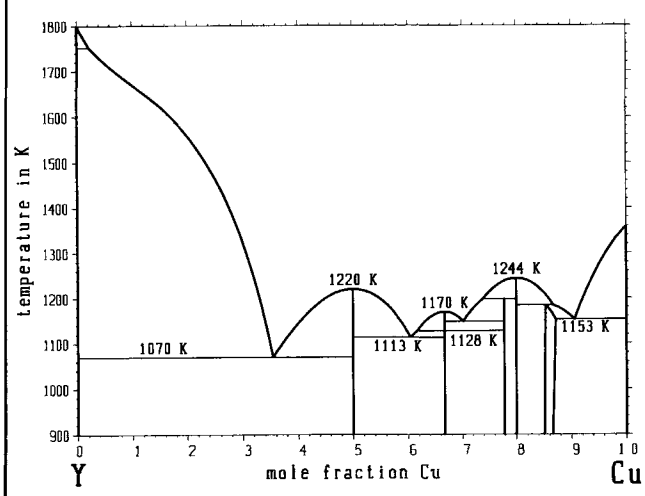

Fig.4 The Y-Cu phase diagram.

dom, nonperiodic defects decreasing the Cu content. An estimate of a range of homogeneity, however, cannot be extracted from the present X-ray data.

Thermodynamic Optimization

Experimental data from the literature and the present results were used as input for the BINGSS [92Luk] optimizer program to obtain an optimized thermodynamic description for all the phases of the Y-Cu system. Tables 2 and 3 summarize all available experimental data.

Models

Each phase was modelled considering the available experimental data.

Reference state. As reference states for the system, the pure solid elements in their stable states at 298.15 K (Stable Element Reference, SER) were chosen. The Gibbs energies of the elements as a function of temperature were compiled by Dinsdale [91Din]. The equations appear in the form:

$$G^{\varphi} - H^{SER} = a + b \cdot T + c \cdot T \cdot \ln T + dT^2 + e/T + f \cdot T^3 + i \cdot T^7 + j \cdot T^{-9}$$

where in different temperature ranges different sets of the coefficients a to j may be used.

Liquid phase. The Gibbs energy of the liquid phase as a function of the concentration was represented by the Redlich-Kister [48Red] formalism because there is no experimental evidence demanding a more sophisticated model:

$$G^{L} - H^{SER} = G^{ref,L} + G^{id,L} + G^{ex,L}$$

with

$$G^{ref,L} = \left[G_Y^{0,L}(T) - H_Y^{SER}(298.15\ K)\right] \cdot X_Y + \left[G_{Cu}^{0,L}(T) - H_{Cu}^{SER}(298.15\ K)\right] \cdot X_{Cu}$$

$$G^{id,L} = RT \cdot \left[X_Y \cdot \ln\left(X_Y\right) + X_{Cu} \cdot \ln\left(X_{Cu}\right)\right]$$

$$G^{ex,L} = X_Y \cdot X_{Cu} \cdot \sum_{i=0}^{n} \left(a_i^L + b_i^L \cdot T\right) \cdot \left(X_Y - X_{Cu}\right)^i \qquad \text{(Eq 1)}$$

In the above equation: G^{φ} is molar Gibbs energy of phase φ, X_Y and X_{Cu} are gross mole fractions of the phase, $H_i^{SER}(298.15)$ is enthalpy of the pure element i in its stable state at the reference temperature 298.15 K, H^{SER} is an abbreviation of $H_Y^{SER}(298.15\ K) \cdot x_Y + H_{Cu}^{SER}(298.15\ K) \cdot x_{Cu}$, $G_Y^{0,L}(T)$ is Gibbs energy of pure liquid Y at temperature T, and $G_{Cu}^{0,L}(T)$ is Gibbs energy of pure liquid Cu at temperature T.

YCu, YCu_2(r), YCu_2(h), Y_2Cu_7 and YCu_4 phases. All these phases were modelled as stoichiometric phases. For YCu, YCu_2, and Y_2Cu_7, no experimental evidence exists for a deviation from stoichiometry. For YCu_4, a small range of homogeneity seems to exist. It is impossible to determine its boundaries by XRD, and no other experimental data are available that provide quantitative information on it.

The Gibbs energy for the stoichiometric phases was given by the formula:

Table 4 Parameter Set for the Y-Cu System

Phase	Model	Parameter	a_ν, J/mol	b_ν, J/mol
Liquid	Redlich-Kister	$L^{L,0}_{Y,Cu}$	–88 958.06	21.56669
		$L^{L,1}_{Y,Cu}$	33 607.31	–3.10462
		$L^{L,2}_{Y,Cu}$	8 083.16	–0.74669
cph	Redlich-Kister	$L^{cph,0}_{Y,Cu}$	(0)	(80.0000)
bcc	Redlich-Kister	$L^{bcc,0}_{Y,Cu}$	(0)	(80.0000)
fcc	Redlich-Kister	$L^{fcc,0}_{Y,Cu}$	(0)	(80.0000)
YCu	Stoichiometric	$G^{YCu} - (G^{cph}_{Y} + G^{fcc}_{Cu})$	–44 616.68	4 12162
YCu_2(r)	Stoichiometric	$G^{YCu_2(r)} - (G^{cph}_{Y} + 2 \cdot G^{fcc}_{Cu})$	–65 324.22	6.03462
YCu_2(h)	Stoichiometric	$G^{YCu_2(h)} - (G^{cph}_{Y} + 2 \cdot G^{fcc}_{Cu})$	–51 788.22	–5.96538
Y_2Cu_7(r)	Stoichiometric	$G^{Y_2Cu_7} - (2 \cdot G^{cph}_{Y} + 7 \cdot G^{fcc}_{Cu})$	–168 519.60	15.56757
YCu_4	Stoichiometric	$G^{YCu_4} - (G^{cph}_{Y} + 4 \cdot G^{fcc}_{Cu})$	–89 328.50	8.25205
YCu_6	Wagner-Schottky $(Y,Cu_2)Cu_5$	$G^{YCu_6}_{Y:Cu} - (G^{cph}_{Y} + 5 \cdot G^{fcc}_{Cu})$	–89 797.58	8.29539
		$G^{YCu_6}_{Cu_2:Cu} - 7 \cdot G^{fcc}_{Cu}$	(35 000.0)	(0)
		$L^{YCu_6,0}_{Y,Cu_2:Cu}$	–34 386 27	21.56669

One mole means one mole of formula units for the intermediate compounds and one mole of atoms for the liquid phase and the solid elements. Coefficients shown in parentheses are estimated and were not optimized. The parameter $0 + 80\ T$ expresses negligible solid solubility of the terminal Y and Cu phases.

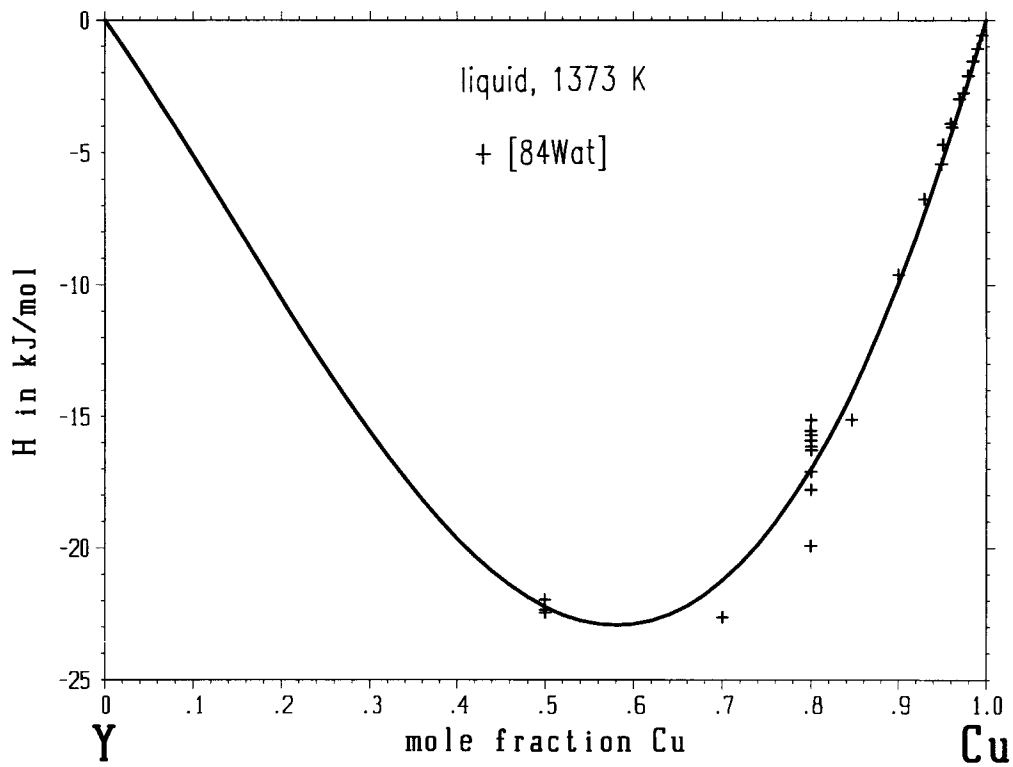

Fig.5 Enthalpy of mixing of Y-Cu liquid.

$$G^{\varphi} - H^{SER} = \left[G^{cph,0}_{Y}(T) - H^{SER}_{Y}(298.15\ K)\right] \cdot X_{Y}$$

$$+ \left[G^{fcc,0}_{Cu}(T) - H^{SER}_{Cu}(298.15\ K)\right] \cdot X_{Cu} + a^{\varphi} + b^{\varphi} \cdot T$$

YCu_6 phase. This phase was modelled using the compound energy formalism [70Hil] with two sublattices, $(Y,Cu_2)(Cu_5)$, where in the first sublattice two atoms of Cu substitute one atom of Y. This structure was determined by [71Bus], and the proposed defect was already suggested in the literature by

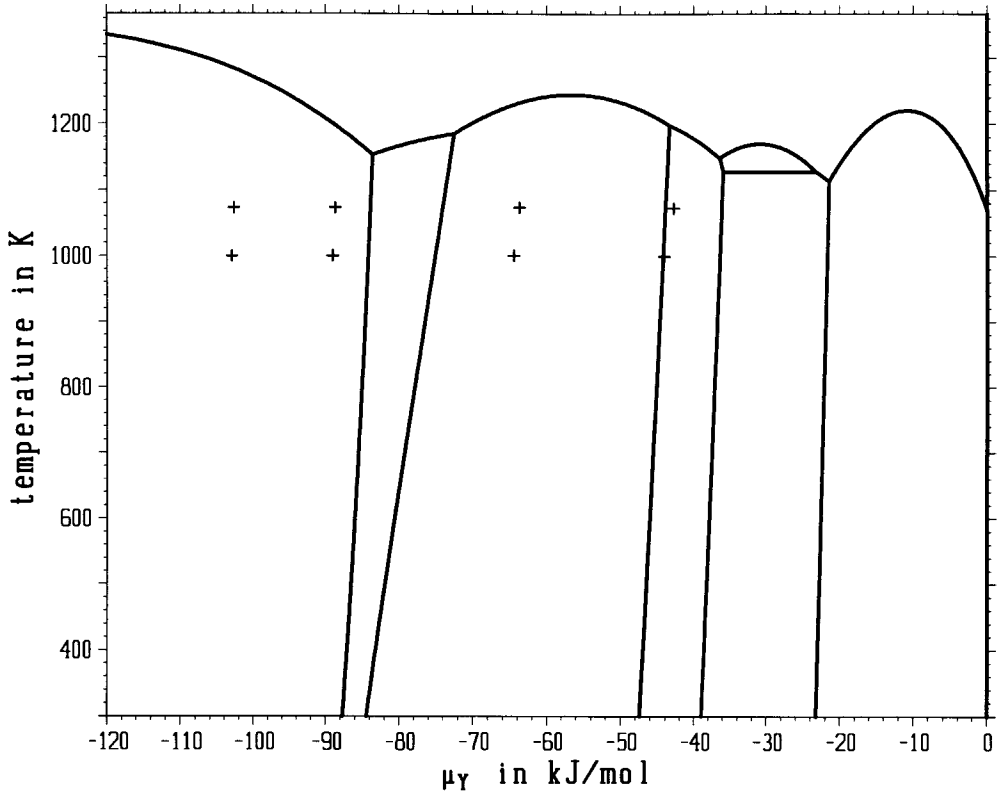

Fig.6 Yttrium chemical potential vs temperature.

[64Sch]. The Gibbs energy, referred to 1 mol of atoms, is given as:

$$G^{\varphi} - H^{SER} = \left(G_Y^{cph,0} - H_Y^{SER}\right) \cdot X_Y$$
$$+ \left(G_{Cu}^{fcc,0} - H_{Cu}^{SER} \cdot X_{Cu} + \left[\left(G_{Cu_2Cu_5}^{\varphi,0} - 7G_{Cu}^{fcc,0}\right) \cdot y_{Cu_2}\right.\right.$$
$$+ \left(G_{Y:Cu_5}^{\varphi,0} - G_Y^{cph,0} - 5G_{Cu}^{fcc,0}\right) \cdot y_Y$$
$$+ R \cdot T\left(y_{Cu_2} \cdot \ln y_{Cu_2} + y_Y \cdot \ln y_Y\right)$$
$$\left.\left. + y_Y \cdot y_{Cu_2} \cdot \sum_{\nu=0}^{n} L_{Y,Cu_2Cu_5}^{\nu} \cdot \left(y_Y - y_{Cu_2}\right)^{\nu}\right] / \left(6 + y_{Cu_2}\right)\right.$$

where y_Y and y_{Cu_2} are site fractions of the species Y and Cu_2 on the first sublattice.

The overall composition $x = X_{Cu}$ and the site fractions are correlated:

$$X = (2 \cdot y_{Cu_2} + 5) / (2 \cdot y_{Cu_2} + y_Y + 5)$$

as the sum of the two site fractions is equal one, $y_Y + y_{Cu_2} = 1$, it follows:

$$X = (2 \cdot y_{Cu_2} + 5) / (y_{Cu_2} + 6)$$

giving

$$5/6 \leq X \leq 1$$

The numerator $2 \cdot y_{Cu_2} + 5$ is the number of Cu atoms ($2 \cdot y_{Cu_2}$ in the first sublattice, 5 in the second one) in one mol of formula units. The denominator $y_{Cu_2} + 6$ is the total number of atoms in one mol of formula units.

Selection of the Adjustable Parameters

For the liquid phase, in Eq 1, n=3 was used. As the phase diagram data and the measured enthalpy of mixing for the liquid are not sufficient to define the composition dependence in the whole range, a constraint between a and b for i=2 and 3 was imposed. Using a factor of 10 850 K, a^{φ} was kept proportional to b^{φ} according to the suggestion of [93Tan]. The same relationship between a and b was applied to the stoichiometric phases because the experimental data are not sufficient to optimize more than one independent coefficient per phase. An attempt to optimize two coefficients per phase results in the calculation of a decomposition of the solid phases at room temperature, for which there is no experimental evidence.

For the YCu_6 phase, the expression $G_{Cu_2:Cu_5}^{0} - 7G_{Cu}^{fcc,0}$, which formally belongs to pure Cu, was fixed to 35 000 J/mol (= 5000 J/mol of atoms). The parameters $G_{Y,Cu_5}^{\varphi,0} - G_Y^{cph,0} - 5G_{Cu}^{fcc,0}$ and $L_{Y,Cu_2:Cu_5}^{\varphi,0}$ are given as linear functions of T and $a + bT$. The two coefficients a and b, however, are set proportional to the factor 10 850 K according to [93Tan]. Terms $L_{Y,Cu_2Cu_5}^{\varphi,\nu}$ with $\nu > 0$ were not used.

The YCu_2(h) phase could not be independently adjusted. Its coefficients were related to those of YCu_2(r). The difference

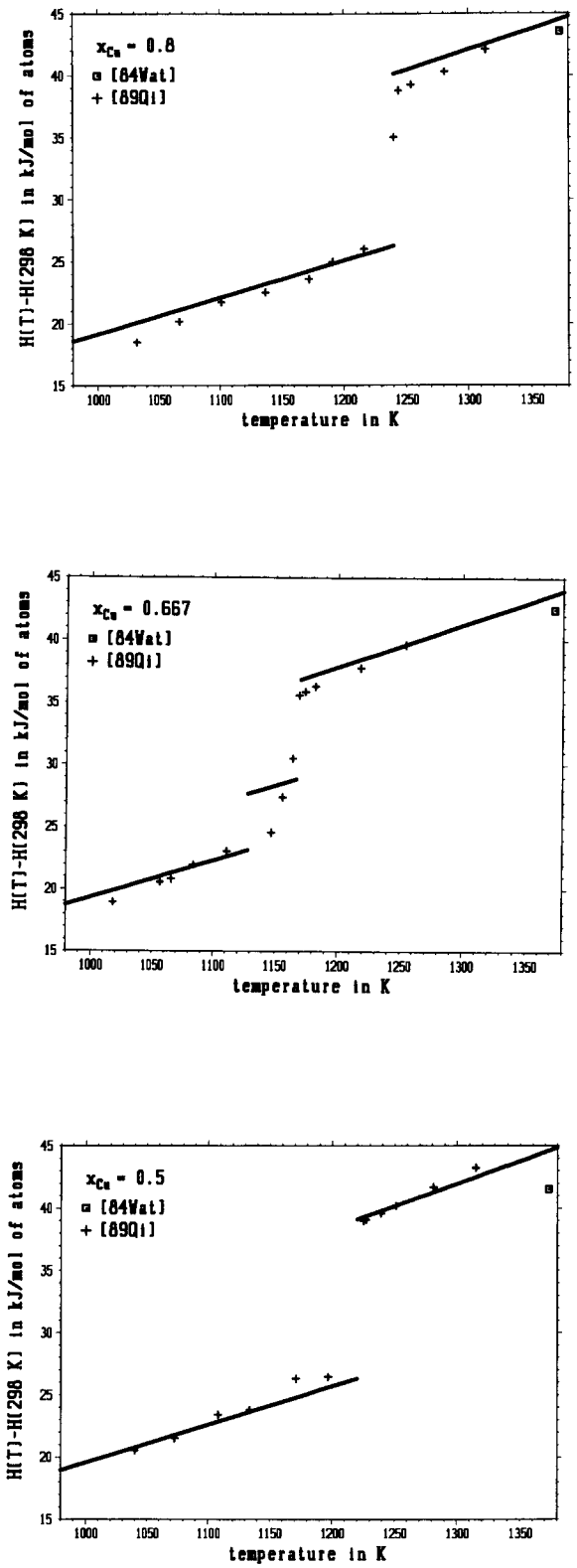

Fig.7 Heat content data of YCu, YCu_2, and YCu_4.

$G^{YCu_2(h)} - G^{YCu_2(r)}$ was fixed to be $(4512 - 4 \cdot T)$ J/mol of atoms. With this constraint, the temperature of the transformation is always calculated at 1128 K (the experimental temperature), and the enthalpy of transformation appears to be about half of the enthalpy of melting. This enthalpy seems to be unusually large, but the ratio of the areas between the DTA peaks of transformation and melting demands this quantity.

Results

Figure 4 shows the phase diagram calculated using the coefficients given in Table 4. Figure 5 presents the experimental data for the enthalpies of mixing of the liquid phase. The calculated function agrees well with the data, which however, cover the Cu-rich part only. In Fig. 6, the Y chemical potential, μ_Y, as a function of temperature is plotted. The experimental data of [89Yan] demand much more negative G values for the YCu_6, YCu_4, and Y_2Cu_7 phases as found in the optimization. Adjusting these data would result in a contradiction to the enthalpies of formation of these phases which are determined by a combination of the enthalpy of mixing of liquid data [84Wat] with the heat content $[H(T) - H(298\ K)]$ data of [89Qi] which include the enthalpies of melting. Because the enthalpy values are considered to be more reliable, the μ values were not used in the optimization; they are in contradiction with the phase diagram. Figure 7 shows the heat content as a function of temperature for the YCu, YCu_2, and YCu_4 compounds.

Conclusions

The Y-Cu binary phase diagram presented in this work is the result of an interactive use of conventional experimental techniques for phase diagram determination with thermodynamic computational optimization methods. The high melting temperature of YCu_2 reported previously in the literature and the invariant temperature of the YCu + liquid + YCu_2 eutectic were shown at the beginning of the optimization to be contradictory data. This fact motivated the experimental investigation, which provided evidence for the existence of a high temperature phase with an associated change of enthalpy during the transformation. The complete experimental characterization of the high temperature phase is a quite difficult problem due to the reactivity of the samples. Consequently, at the present time, the calculated phase diagram represents is the best information available for this system.

The solubility range of the YCu_4 phase still remains to be determined.

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